

1.Student Details :

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About Me : I am a Diploma level student at IIT Madras BS degree with a deep interest in learning about web applications and web application development to solve real world problems. I enjoy building meaningful applications that combine learning, analytics and user experience.

2.Project Details:

Project Title : PLACEMENT PORTAL APPLICATION

Problem Statement :

- To design and build a web based application that connects Students, Companies and the College, enabling a seamless and efficient recruitment process.

Approach :

- The app was built using Flask as the backend framework with a modular structure. It allows students to login, see company drives and apply for drives and allows companies to create placement drives.

3. AI/LLM Declaration :

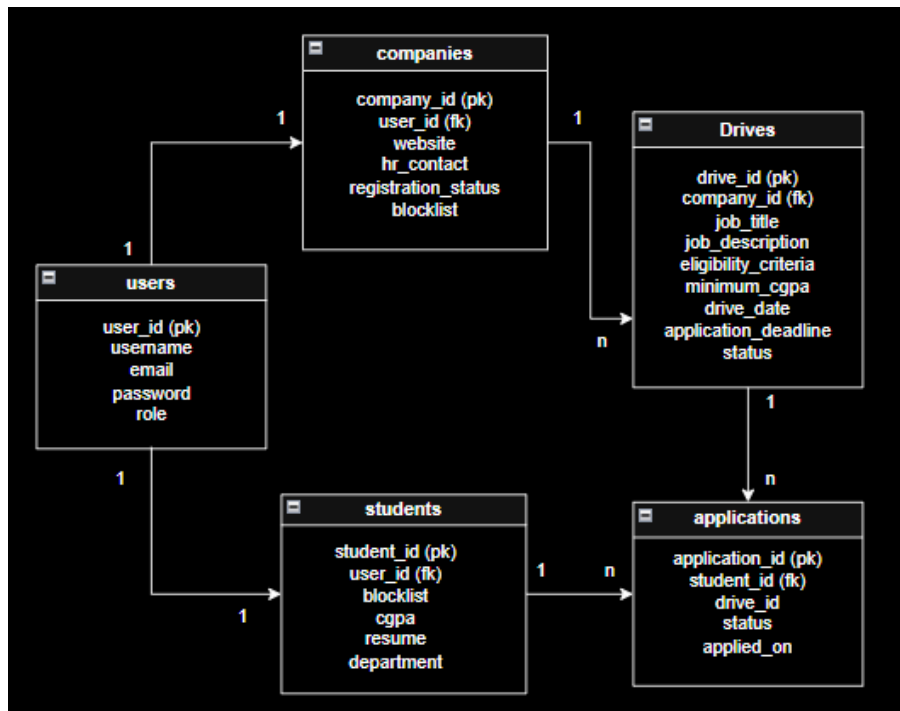
- I used ChatGpt(GPT-5) to assist in writing frontend design code (html, css, jinja) , creating API documentation samples and improving variable naming consistency. The extend of AI/LLM usage is around 10-15%, limited to code suggestions and documentation formatting. All final implementation logic, debugging , and interaction were done manually.

4.Technologies and Frameworks Used

Technology / Library	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages

Bootstrap	Frontend styling and responsive design
SQLite	Lightweight local database for storing user data
werkzeug	provides utilities for building web applications, including request handling, routing, and secure file processing

5. ER Diagram



6.API Resource Endpoint:

1.Common Routes

Endpoint	Method	Description
/	GET	Renders the home page
/login	GET, POST	Handles users login and redirects based on role
/logout	GET	Logs out the user by clearing the session
/register/student	GET,POST	Register a new student
/register/company	GET, POST	Registers new company (pending)

2. Admin Module

Endpoint	Method	Description
admin/adminDashboard	GET	View admin dashboard

admin/students	GET	View all students
/admin/block_student/<id>	GET	Block a student
/admin/unblock_student/<id>	GET	Unblock a student
/admin/delete_student/<id>	GET	Delete a student
/admin/pendingApproval	GET	View pending company approvals
/company/approve/<id>	GET	Approve company registration
/admin/pending-drives	GET	View pending placement drives
/admin/approve-drive/<id>	GET	Approve drive
/admin/reject-drive/<id>	GET	Reject drive
/admin/ongoingDrives	GET	View ongoing drives
/admin/delete_drive/<id>	GET	Delete drive
/admin/applications	GET	View all student applications
/admin/drivesHistory	GET	View all drives (ongoing + completed)
/admin/drive/<drive_id>/applications	GET	View applications for a specific drive
/admin/drive/<drive_id>/selected	GET	View selected students for a drive
/admin/search	GET	Search students/companies by name or ID
/admin/delete_company/<id>	GET	Delete company
/admin/block_company/<id>	GET	Block company
/admin/unblock_company/<id>	GET	Unblock company
/admin/company/<id>/drives	GET	View drives of a specific company

3. Student Module

Endpoint	Method	Description
/student/dashboard	GET	View student dashboard (drives + applications)
/apply/<drive_id>	GET	Apply for a placement drive
/student/profile	GET	View student profile
/student/profile	GET	Update student profile (department, CGPA, resume upload)

4. Company Module

Endpoint	Method	Description
/company/companyDashboard	GET	View company dashboard (profile + drives)

/company/createDrives	GET, POST	Create a new placement drive
/company/edit_drive/<id>	GET, POST	Edit an existing drive
/company/delete_drive/<id>	GET	Delete a drive
/company/applications/<drive_id>	GET	View applications for a specific drive
/company/application/update/<id>/<status>	GET	Update application status (selected/rejected)

7. Architecture and Features

Architecture Overview:

- **main.py** – Entry point that initializes the app
- **/applications/controllers** – Contains controllers where the logics and api endpoints are created.
- **/applications/models.py** – database models using SQLAlchemy
- **/db_directory** – Contains .db file which is automatically created when running main.py
- **/static** – css and resume files
- **/templates** – jinja2, HTML templates

Features of this application :

- Role-Based Access Control
- Placement drive management
- Student application system
- Resume Upload and Viewing
- Admin approval system
- Application Status management

8. Video Presentation